Advanced Computer-Aided VLSI System Design
Lab3: Automatic Test Pattern Generation (ATPG)

TA: Can-Lin Wu r13943017@ntu.edu.tw

Objectives

In this lab, you will learn how to generate single stuck-at fault test patterns for our simple ALU.

Data Preparation

1. Create a work directory and copy the lab files into it
2. Check if you have these files:

Filename	Description
ALU_syn_dft.v	Scan-ready gate level netlist for the simple ALU
ALU_syn_dft.spf	STIL format test protocol file (generated by DFT Compiler)

3. Check the contents of these files.

Invoke Tetramax

To invoke Tetramax, first include the following C shell script

% **cb**

And then, you can do either one

% **tmax -nogui** (command mode)

% **tmax &** (GUI mode)

In this Lab, we will use the command mode. You are welcome to try the GUI mode. In the GUI mode, you can type in your commands in the command line, located at the bottom of the window.

STEP 1: Read Input Files

Read in your design and cell library

% **read_netlist ALU_syn_dft.v**

Read in cell library in Verilog format

% **read_netlist <cell.v>**

NOTE: unlike the design compiler, Tetramax does not take the library in .ddc format. Instead, Tetramax reads library in Verilog format.

The location of N16ADFP_StdCell.v is at:

/home/cad/DAFP0203001_2_X/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/VERILOG/N16ADFP_StdCell.v

STEP 2: Build Model

1. This step builds a model of your design for ATPG.
   % **run_build_model ALU**

2. To probe further for the violations, type
   % **report_violations -all**
   Examine these violations,

STEP 3: Run DRC

The spf file tells the ATPG how to operate the circuit in test mode.
This step run design rule checking. Please type,
% **run_drc ALU_syn_dft.spf**

Question: Does our design pass all rules? _____

STEP 4: Set fault list

Specify the fault model. We use stuck-at fault model in this lab.
% **set_faults -model stuck**
Inject all faults into the circuit under test
% **add_faults -all**

STEP 5: Run ATPG

To obtain the maximum test coverage while minimizing the number of patterns, we can set the ATPG options as follows.
% **set_atpg -abort_limit 100 -merge high**
% **run_atpg -auto_compression**

Question: How many total faults do we have (Uncollapsed)? _____

STEP 5 Review Results

To get a summary of your ATPG results, type
% **report_summaries**
Question 1: How many faults are detected? _____
Question 2: What is the test coverage? _____
Question 3: How many test patterns do we have? _____

The default test coverage is obtained by using uncollapsed fault list. To use the collapsed fault list, type

% set_faults -report collapsed

% report_summaries

TetraMAX ATPG defines test coverage and fault coverage as follows:

$$\text{Test Coverage} = \frac{\text{DT} + (\text{PT} \times \text{PT_credit})}{\text{All_Faults} - \text{UD} - (\text{AN} \times \text{AU_credit})} \times 100$$

$$\text{Fault Coverage} = \frac{\text{DT} + (\text{PT} \times \text{PT_credit})}{\text{All_Faults}} \times 100$$

To display fault coverage in addition to test coverage with the summary report type

% set_faults -fault_coverage

% report_summaries

STEP 7 Save Patterns

To see your test patterns, type

% report_patterns -all

Save your patterns into files. There are many formats supported by TetraMAX. WGL (Waveform Generation Language), and STIL (Standard Test Interface Language) are two of the popular formats. Depends on the test equipment you use, you can choose other different formats. Although ATPG should give us "correct" patterns, it is always good to verify the patterns. TetraMax can also generate script shell script for either NCVerilog or VCS to simulate patterns. Use –nc or –vcs option to generate shell script for the STIL format patterns.

% write_patterns ALU_syn_ATPG.wgl -format WGL

% write_patterns ALU_syn_ATPG.stil -format STIL -vcs

Question: How many files do you get by typing the last command? _____

Try to run Verilog simulations and see if you get correct results. Exit tmax and run the following command in Linux to generate testbench.

% bash

% export SYNOPSYS_TMAX=/share1/cad/synopsys/tmax/2022.03

% stil2verilog ALU_syn_ATPG.stil ALU_syn_ATPG_stildpv

Open ALU_syn_ATPG_vcs.sh

1. Add "STILDPV_HOME=/share1/cad/synopsys/tmax/2022.03/linux64/stildpv"
2. Set LIB file to /share1/tech/ADFP/Executable_Package/Collaterals/IP/stdcell/N16ADFP_StdCell/VERILOG/N16ADFP_StdCell.v "

3. Netlist="ALU_syn_dft.v"
4. Edit SIMULATOR="vcs" to SIMULATOR="vcs -full64"

% bash ALU_syn_ATPG_vcs.sh

Checkpoints

Please check with TAs before leaving this lab to make sure the following goals are accomplished and to get credits.

1. Show your ATPG simulations without any mismatch.

```
make: warning:  Clock skew detected.  Your build may be incomplete.
Chronologic VCS simulator copyright 1991-2022
Contains Synopsys proprietary information.
Compiler version T-2022.06_Full64; Runtime version T-2022.06_Full64;  Apr  7 13:17 2026
////////////////////////////////////////////////////////////////////////////
 MAX TB Version T-2022.03
 Test Protocol File generated from original file "ALU_syn_ATPG.stil"
 STIL file version: 1.0
////////////////////////////////////////////////////////////////////////////

XTB: Starting serial simulation of 28 patterns
XTB: Begin serial scan load for pattern 0 (T=200.00 ns, V=3)
XTB: Begin serial scan load for pattern 5 (T=15200.00 ns, V=153)
XTB: Begin serial scan load for pattern 10 (T=30200.00 ns, V=303)
XTB: Begin serial scan load for pattern 15 (T=45200.00 ns, V=453)
XTB: Begin serial scan load for pattern 20 (T=60200.00 ns, V=603)
XTB: Begin serial scan load for pattern 25 (T=75200.00 ns, V=753)
XTB: Simulation of 28 patterns completed with 0 mismatches (time: 87100.00 ns, cycles: 871)

          V C S   S i m u l a t i o n   R e p o r t
Time: 87100000 ps
CPU time: 2.878 seconds to compile + .317 seconds to elab + .716 seconds to link + .227 seconds in simulation
```

2. Answer the questions in this lab.